

Day 5: Network Testing Tools

Task 1 — Ping Test

I used the ping command to test connectivity to YouTube.

Command:

```
ping www.youtube.com
```

The command sends packets to the YouTube server and shows the response time.

Observation:

The ping command helped me check whether my computer could reach the destination and how much time the response took.

Note: Enter the average value shown in your own command output.

Task 2 — Traceroute Test

On Windows, the traceroute command is called
tracert.

Command:

tracert www.flipkart.com

The command shows the different hops between my
computer and the destination.

Result:

Hop No.	IP Address / Host
1	_____
2	_____
3	_____
4	_____
5	_____
...	...

Observation:

Each hop represents a network device or router
through which the packets travel. Some hops may
show * * * because the device may not respond to
traceroute requests.

Note: Copy the hops from your own tracert output rather than using fixed example values, because the route can be different for different networks.

Task 3 — nslookup

I used the nslookup command to find the IP address associated with Zomato.

Command:

```
nslookup www.zomato.com
```

Result:

IP Address: _____

Observation:

nslookup queries the DNS system and shows the IP address associated with a domain name.

Zomato may return more than one IP address, depending on DNS and the network being used. Record the address shown on your computer.

Task 4 — Ping a Local Device

I used the ping command to test connectivity to another device on my local network.

For example:

```
ping 192.168.10.11
```

If the device is online:

The computer receives replies such as:

```
Reply from 192.168.10.11: bytes=32 time<1ms  
TTL=128
```

This means the device is reachable on the local network.

If the device is offline:

The ping command may show:

```
Request timed out.
```

This means that my computer did not receive a reply from the device. The device may be switched off, disconnected from Wi-Fi, or configured not to respond to ping.

Result:

Ping is useful for quickly checking whether another device is reachable.

Task 5 — Short Report

Network Testing Tools Report

Ping is useful for checking whether a website or device is reachable and for measuring response time. In my YouTube test, I used ping www.youtube.com to check connectivity and response time. Traceroute is useful when I want to find where a connection is taking a different or slow route, so I used tracert www.flipkart.com to view the intermediate hops. Nslookup is useful when I suspect a DNS problem because it checks which IP address belongs to a domain name. I used nslookup www.zomato.com to check its DNS information. These three tools help diagnose different types of network problems.

Simple Summary

Problem	Useful Tool	Why
Website not loading	Ping	Checks basic connectivity
Slow connection	Traceroute	Shows the route and intermediate hops
Suspected DNS issue	nslookup	Checks domain-to-IP resolution

Commands Used

ping www.youtube.com

tracert www.flipkart.com

nslookup www.zomato.com

ping <local-device-IP>